

$$\text{a)} \quad \lim_{x \rightarrow 2} (x^2 - 4)$$

$$\text{j)} \quad \lim_{x \rightarrow 2} \frac{x-2}{x^2-3x+2}$$

$$\text{s)} \quad \lim_{x \rightarrow -2} \frac{x^3+3x^2+2x}{x^2-x-6}$$

$$\text{b)} \quad \lim_{x \rightarrow 0} \frac{x^3-4x}{2x^2+3x}$$

$$\text{k)} \quad \lim_{x \rightarrow 0} \frac{3x+2x^{-1}}{x+4x^{-1}}$$

$$\text{t)} \quad \lim_{x \rightarrow 1} \frac{x^2-2x+1}{x^3-x}$$

$$\text{c)} \quad \lim_{x \rightarrow -1} \frac{x^3}{(x+1)^2}$$

$$\text{l)} \quad \lim_{x \rightarrow 2} \frac{x^2-3x+2}{x^2-2x}$$

$$\text{u)} \quad \lim_{x \rightarrow 4} \frac{x^2+7x-44}{x^2-6x+8}$$

$$\text{d)} \quad \lim_{x \rightarrow -1} \frac{(x+1)^2(x-1)}{x^3+1}$$

$$\text{m)} \quad \lim_{x \rightarrow 1} \left( \frac{1}{1-x} - \frac{3}{1-x^3} \right)$$

$$\text{v)} \quad \lim_{x \rightarrow 1} \frac{x^2-4}{x^2-3x+2}$$

$$\text{e)} \quad \lim_{x \rightarrow 0} \frac{x^3-2x^2+x}{2x^3+x^2-2x}$$

$$\text{n)} \quad \lim_{x \rightarrow 2} \frac{x^2-x-2}{x^2-2x}$$

$$\text{w)} \quad \lim_{x \rightarrow 1} \frac{x^3-5x+4}{x^3-1}$$

$$\text{f)} \quad \lim_{x \rightarrow 1} \frac{x^2+2x+3}{(x-1)^2}$$

$$\text{o)} \quad \lim_{x \rightarrow 2} \frac{x^2+5}{x^2-3}$$

$$\text{x)} \quad \lim_{x \rightarrow 2} \frac{x^2-4}{x-2}$$

$$\text{g)} \quad \lim_{x \rightarrow 0} \frac{x^4-4x^3+x^2}{x^3+x^2+x}$$

$$\text{p)} \quad \lim_{x \rightarrow 1} \frac{3x^4-4x^3+1}{(x-1)^2}$$

$$\text{y)} \quad \lim_{x \rightarrow 2} \frac{x^2-4}{x^2-3x+2}$$

$$\text{h)} \quad \lim_{x \rightarrow -1} \frac{x^3+x^2+x+1}{x^4+x^2-2}$$

$$\text{q)} \quad \lim_{x \rightarrow -2} \frac{3x+6}{x^3+8}$$

$$\text{z)} \quad \lim_{x \rightarrow 1} \left( \frac{1}{x^2-1} - \frac{2}{x^4-1} \right)$$

$$\text{i)} \quad \lim_{x \rightarrow 2} \frac{(x+1)^2}{2-x}$$

$$\text{r)} \quad \lim_{x \rightarrow 2} \frac{x+1}{x-1}$$

$$\text{Z)} \quad \lim_{x \rightarrow 3} \frac{x-3}{x^2-5x+6}$$

$$\text{a)} \quad \lim_{x \rightarrow \infty} \frac{x^2 - 1}{2x^2 + 1}$$

$$\text{j)} \quad \lim_{x \rightarrow -\infty} \frac{x^3 + x^4 - 1}{2x^5 + x - x^2}$$

$$\text{s)} \quad \lim_{x \rightarrow \infty} \frac{(x+3)(x+4)(x+5)}{x^4 + x - 11}$$

$$\text{b)} \quad \lim_{x \rightarrow -\infty} \frac{x^3 + x^2 - 4}{2x^3 + x + 11}$$

$$\text{k)} \quad \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + 1} + x)^2}{\sqrt[3]{x^6 + 1}}$$

$$\text{t)} \quad \lim_{x \rightarrow -\infty} \frac{8x - 2x^5 + x^6}{11x + 5x^3 + 3x^5}$$

$$\text{c)} \quad \lim_{x \rightarrow \infty} \frac{3x^2 + 2x - 1}{x^3 - x + 2}$$

$$\text{l)} \quad \lim_{x \rightarrow -\infty} \frac{x^6 + 7x^4 - 40}{1 - x - 5x^7}$$

$$\text{u)} \quad \lim_{x \rightarrow \infty} \left( \frac{x^3}{2x^2 - 1} - \frac{x^2}{2x + 1} \right)$$

$$\text{d)} \quad \lim_{x \rightarrow \infty} \left( \frac{x^3}{x^2 + 2} - x \right)$$

$$\text{m)} \quad \lim_{x \rightarrow \infty} \frac{(x+1)(x-2)}{3x^2 + 6x - 5}$$

$$\text{v)} \quad \lim_{x \rightarrow \infty} \left( x^2 - \frac{x^4 - 1}{x^2 - 2} \right)$$

$$\text{e)} \quad \lim_{x \rightarrow \infty} \frac{x^2 + 3x - 4}{3x^2 - 2x + 5}$$

$$\text{n)} \quad \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 1}}{x}$$

$$\text{w)} \quad \lim_{x \rightarrow \infty} \frac{(x-1)^{100}(6x+1)^{200}}{(3x+5)^{300}}$$

$$\text{f)} \quad \lim_{x \rightarrow \infty} \frac{x(x-1)(x-2)}{x^2 + 6x - 9}$$

$$\text{o)} \quad \lim_{x \rightarrow \infty} \left( \frac{3x^2 + 2x + 1}{x^2 - 3x + 2} \right)^4$$

$$\text{x)} \quad \lim_{x \rightarrow \infty} \frac{\sqrt[4]{x^5} + \sqrt[5]{x^3} + \sqrt[6]{x^8}}{\sqrt[3]{x^4 + 2}}$$

$$\text{g)} \quad \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 9}}{x + 3}$$

$$\text{p)} \quad \lim_{x \rightarrow -\infty} \frac{5x^3 - x^2 + x}{1 - x - 3x^2}$$

$$\text{y)} \quad \lim_{x \rightarrow -\infty} \frac{x^2(2x+1)(3x-2)}{2x^2(5x-8)(x+6)}$$

$$\text{h)} \quad \lim_{x \rightarrow \infty} \left( \frac{x^2 + x - 1}{2x^2 - x + 1} \right)^3$$

$$\text{q)} \quad \lim_{x \rightarrow \infty} \frac{1 + x - 3x^3}{1 + x^2 + 3x^3}$$

$$\text{z)} \quad \lim_{x \rightarrow \infty} \left( \frac{2x^8 + 8x^6 + 6x^4}{4x^8 - x^6 + 12x^4} \right)^5$$

$$\text{i)} \quad \lim_{x \rightarrow \infty} \frac{x^2 + 2x + 1}{5x}$$

$$\text{r)} \quad \lim_{x \rightarrow -\infty} \left( \frac{x^3 - 8}{x^4 + 16} \right)^{10}$$

$$\text{Z)} \quad \lim_{x \rightarrow -\infty} \frac{(2x-3)^{20}(3x+2)^{30}}{(2x+1)^{50}}$$

|           |                                                                |           |                                                             |           |                                                                                                     |
|-----------|----------------------------------------------------------------|-----------|-------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------|
| <b>a)</b> | $\lim_{x \rightarrow 0} \frac{\sqrt{1+2x}-1}{3x}$              | <b>j)</b> | $\lim_{x \rightarrow \infty} (\sqrt{x-2}-\sqrt{x})$         | <b>s)</b> | $\lim_{x \rightarrow \infty} \frac{\sqrt{x+2}-\sqrt{2}}{x}$                                         |
| <b>b)</b> | $\lim_{x \rightarrow 0} \frac{\sqrt{1+x}-\sqrt{1-x}}{x}$       | <b>k)</b> | $\lim_{x \rightarrow \infty} (\sqrt{x^2+x}-x)$              | <b>t)</b> | $\lim_{x \rightarrow \infty} \frac{\sqrt{x+5}-\sqrt{5}}{\sqrt{x}-5}$                                |
| <b>c)</b> | $\lim_{x \rightarrow 0} \frac{x-\sqrt{x}}{\sqrt{x}}$           | <b>l)</b> | $\lim_{x \rightarrow \infty} (\sqrt{x^2+x}-x)$              | <b>u)</b> | $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2+9}-\sqrt{x^2-9}}{6x}$                                  |
| <b>d)</b> | $\lim_{x \rightarrow 5} \frac{\sqrt{x-1}-2}{x^2-25}$           | <b>m)</b> | $\lim_{x \rightarrow \infty} (\sqrt{x-3}-\sqrt{x})$         | <b>v)</b> | $\lim_{x \rightarrow \infty} \frac{\sqrt{x-1}-2x}{x-7}$                                             |
| <b>e)</b> | $\lim_{x \rightarrow 3} \frac{\sqrt{x+6}-3}{x^3-5x^2+3x+9}$    | <b>n)</b> | $\lim_{x \rightarrow \infty} \sqrt{x}(\sqrt{x-3}-\sqrt{x})$ | <b>w)</b> | $\lim_{x \rightarrow \infty} \frac{\sqrt{x}-6x}{3x+1}$                                              |
| <b>f)</b> | $\lim_{x \rightarrow 9} \frac{3-\sqrt{x}}{27-\sqrt{x^3}}$      | <b>o)</b> | $\lim_{x \rightarrow \infty} x(\sqrt{x^2+1}-x)$             | <b>x)</b> | $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2+1}+\sqrt{x}}{\sqrt[4]{x^3+x}-x}$                       |
| <b>g)</b> | $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x}-\sqrt[3]{1-x}}{x}$ | <b>p)</b> | $\lim_{x \rightarrow \infty} x(\sqrt{x^2+1}-x)$             | <b>y)</b> | $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2+1}+\sqrt{x}}{\sqrt[4]{x^2+1}-x}$                       |
| <b>h)</b> | $\lim_{x \rightarrow 1} \frac{x^{2/3}-1}{x^{3/5}-1}$           | <b>q)</b> | $\lim_{x \rightarrow \infty} (\sqrt{x^2+1}-x)$              | <b>z)</b> | $\lim_{x \rightarrow \infty} \frac{\sqrt[3]{x}-2\sqrt{x^3}}{\sqrt[4]{x^5+x}\sqrt{x}}$               |
| <b>i)</b> | $\lim_{x \rightarrow 1} \frac{1-\sqrt[n]{x}}{1-\sqrt[m]{x}}$   | <b>r)</b> | $\lim_{x \rightarrow \infty} (\sqrt{x^2+1}-x)$              | <b>Z)</b> | $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2+1}-\sqrt[3]{x^2+1}}{2\sqrt[4]{x^4+1}-\sqrt[5]{x^4+1}}$ |

$$\text{a)} \quad \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^{3x}$$

$$\text{j)} \quad \lim_{x \rightarrow \infty} \left(\frac{7x+10}{1+7x}\right)^{\frac{x}{3}}$$

$$\text{s)} \quad \lim_{x \rightarrow \infty} \left(\frac{2x-5}{2x-2}\right)^{4x^2}$$

$$\text{b)} \quad \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x^2}\right)^{3x-4}$$

$$\text{k)} \quad \lim_{x \rightarrow \infty} \left(\frac{2x-1}{2x+1}\right)^x$$

$$\text{t)} \quad \lim_{x \rightarrow \infty} \left(\frac{3x+6}{3x-1}\right)^{x^2}$$

$$\text{c)} \quad \lim_{x \rightarrow \infty} \left(1 + \frac{1}{5x}\right)^{2x+6}$$

$$\text{l)} \quad \lim_{x \rightarrow \infty} \left(\frac{6+4x}{2+4x}\right)^{3-2x}$$

$$\text{u)} \quad \lim_{x \rightarrow \infty} \left(\frac{x^2+2x+2}{x^2+3}\right)^x$$

$$\text{d)} \quad \lim_{x \rightarrow \infty} \left(1 + \frac{7}{3x}\right)^{x-1}$$

$$\text{m)} \quad \lim_{x \rightarrow \infty} \left(\frac{x+5}{x+4}\right)^{2x-1}$$

$$\text{v)} \quad \lim_{x \rightarrow 0} (1+2x)^{\frac{1}{x}}$$

$$\text{e)} \quad \lim_{x \rightarrow \infty} \left(1 - \frac{1}{3x}\right)^x$$

$$\text{n)} \quad \lim_{x \rightarrow \infty} \left(\frac{2x+5}{2x}\right)^{3x-7}$$

$$\text{w)} \quad \lim_{x \rightarrow 0} (1+5x)^{\frac{1}{8x}}$$

$$\text{f)} \quad \lim_{x \rightarrow \infty} \left(1 - \frac{5}{x}\right)^x$$

$$\text{o)} \quad \lim_{x \rightarrow \infty} \left(\frac{2x+1}{2x-3}\right)^{3x}$$

$$\text{x)} \quad \lim_{x \rightarrow \infty} \left[ \ln \left( \frac{3x+1}{3x-5} \right)^{2-x} \right]$$

$$\text{g)} \quad \lim_{x \rightarrow \infty} \left(\frac{x-1}{x+1}\right)^x$$

$$\text{p)} \quad \lim_{x \rightarrow \infty} \left(\frac{x-1}{x+3}\right)^{5-4x}$$

$$\text{y)} \quad \lim_{x \rightarrow \infty} x [\ln(x+3) - \ln x]$$

$$\text{h)} \quad \lim_{x \rightarrow \infty} \left(\frac{x+6}{x+5}\right)^x$$

$$\text{q)} \quad \lim_{x \rightarrow \infty} \left(\frac{x}{x+1}\right)^x$$

$$\text{z)} \quad \lim_{x \rightarrow \infty} x [\ln x - \ln(x+2)]$$

$$\text{i)} \quad \lim_{x \rightarrow \infty} \left(\frac{x-3}{x}\right)^{\frac{x}{2}}$$

$$\text{r)} \quad \lim_{x \rightarrow \infty} \left(\frac{3x-2}{3x+1}\right)^{2x}$$

$$\text{Z)} \quad \lim_{x \rightarrow \infty} (x+1) [\ln(x+1) - \ln x]$$

$$\text{a)} \quad \lim_{x \rightarrow 0} \frac{\sin 10x}{10x}$$

$$\text{b)} \quad \lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$$

$$\text{c)} \quad \lim_{x \rightarrow 0} \frac{\tan 8x}{x}$$

$$\text{d)} \quad \lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x}$$

$$\text{e)} \quad \lim_{x \rightarrow 0} \frac{\tan 5x}{\sin 4x}$$

$$\text{f)} \quad \lim_{x \rightarrow 0} \frac{\tan 5x}{\tan 6x}$$

$$\text{g)} \quad \lim_{x \rightarrow 0} \frac{\tan x}{3x}$$

$$\text{h)} \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$$

$$\text{i)} \quad \lim_{x \rightarrow 0} \frac{x}{\sqrt{1 - \cos x}}$$

$$\text{j)} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x^3}$$

$$\text{k)} \quad \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x \sin x}$$

$$\text{l)} \quad \lim_{x \rightarrow 0} \frac{\sin^3(x/2)}{x^3}$$

$$\text{m)} \quad \lim_{x \rightarrow 0} \frac{\sin 4x + \sin 7x}{\sin 3x}$$

$$\text{n)} \quad \lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos x^2}}{1 - \cos x}$$

$$\text{o)} \quad \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$$

$$\text{p)} \quad \lim_{x \rightarrow \infty} x \sin \frac{\pi}{x}$$

$$\text{q)} \quad \lim_{x \rightarrow -1} \frac{x^3 + 1}{\sin(x + 1)}$$

$$\text{r)} \quad \lim_{x \rightarrow 0} \frac{|\sin x|}{x}$$

$$\text{s)} \quad \lim_{x \rightarrow 1} \frac{\tan(x - 1)}{\sqrt{x} - 1}$$

$$\text{t)} \quad \lim_{x \rightarrow 0} \frac{\sqrt{\cos x} - 1}{\sin^2 x}$$

$$\text{u)} \quad \lim_{x \rightarrow 0} \frac{\sin 3x + \sin 5x}{\sin 2x}$$

$$\text{v)} \quad \lim_{x \rightarrow 0} \frac{\cos x - \cos^3 x}{x^2}$$

$$\text{w)} \quad \lim_{x \rightarrow 0} x \cot 2x$$

$$\text{x)} \quad \lim_{x \rightarrow 0} \sqrt{\frac{1 - \cos x}{x^2}}$$

$$\text{y)} \quad \lim_{x \rightarrow \pi} \frac{\sqrt{1 - \tan x} - \sqrt{1 + \tan x}}{\sin 2x}$$

$$\text{z)} \quad \lim_{x \rightarrow 0} \frac{\ln x}{\ln(\sin x)}$$

$$\text{Z)} \quad \lim_{x \rightarrow 0} \frac{\ln(\sin 2x)}{\ln(\sin x)}$$

$$\text{a)} \quad \lim_{x \rightarrow 0} \frac{3^x - 1}{6^x - 1}$$

$$\text{j)} \quad \lim_{x \rightarrow -\infty} \left( \ln \frac{x-1}{x+1} \right)$$

$$\text{s)} \quad \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{\cos 2x}$$

$$\text{b)} \quad \lim_{x \rightarrow \infty} x(2^{\frac{1}{x}} - 1)$$

$$\text{k)} \quad \lim_{x \rightarrow 0} \frac{1}{x} \ln \sqrt{\frac{1+x}{1-x}}$$

$$\text{t)} \quad \lim_{x \rightarrow \infty} (\sin \sqrt{x+1} - \sin \sqrt{x})$$

$$\text{c)} \quad \lim_{x \rightarrow 0} (e^x + x)^{\frac{1}{x}}$$

$$\text{l)} \quad \lim_{x \rightarrow e} \frac{\ln x^x - x}{2 - \ln x^2}$$

$$\text{u)} \quad \lim_{x \rightarrow \pi/4} (\tan x)^{\tan 2x}$$

$$\text{d)} \quad \lim_{x \rightarrow \infty} \frac{e^{\frac{1+x}{1-x}}}{2}$$

$$\text{m)} \quad \lim_{x \rightarrow 0} \frac{1 - 3^x}{\sin 3x}$$

$$\text{v)} \quad \lim_{x \rightarrow 0} \frac{\arcsin x}{x}$$

$$\text{e)} \quad \lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{2}$$

$$\text{n)} \quad \lim_{x \rightarrow -\infty} 2^x \sin 2\pi x$$

$$\text{w)} \quad \lim_{x \rightarrow 0} \frac{\arcsin 5x}{3x}$$

$$\text{f)} \quad \lim_{x \rightarrow -\infty} \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

$$\text{o)} \quad \lim_{x \rightarrow 0} \frac{\sin 2x}{\sqrt{x+3} - \sqrt{3}}$$

$$\text{x)} \quad \lim_{x \rightarrow -\infty} \frac{e^{-x}}{\arcsin(1/x)}$$

$$\text{g)} \quad \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$$

$$\text{p)} \quad \lim_{x \rightarrow \infty} x^2 \left( 1 - \cos \frac{1}{x} \right)$$

$$\text{y)} \quad \lim_{x \rightarrow \infty} \arccos(\sqrt{x^2 + x} - x)$$

$$\text{h)} \quad \lim_{x \rightarrow 0} \frac{\ln(1+3x)}{x}$$

$$\text{q)} \quad \lim_{x \rightarrow 0} (\cos x)^{\frac{1}{x}}$$

$$\text{z)} \quad \lim_{x \rightarrow \infty} \frac{\operatorname{arccot} x}{x^2 - x}$$

$$\text{i)} \quad \lim_{x \rightarrow e} \frac{\ln x - 1}{x - e}$$

$$\text{r)} \quad \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{\pi - 2x}$$

$$\text{Z)} \quad \lim_{x \rightarrow 1} \sqrt{\frac{\pi}{2} - \arctan \frac{1}{(x-1)^2}}$$